

Monitoring Intrusion Detection



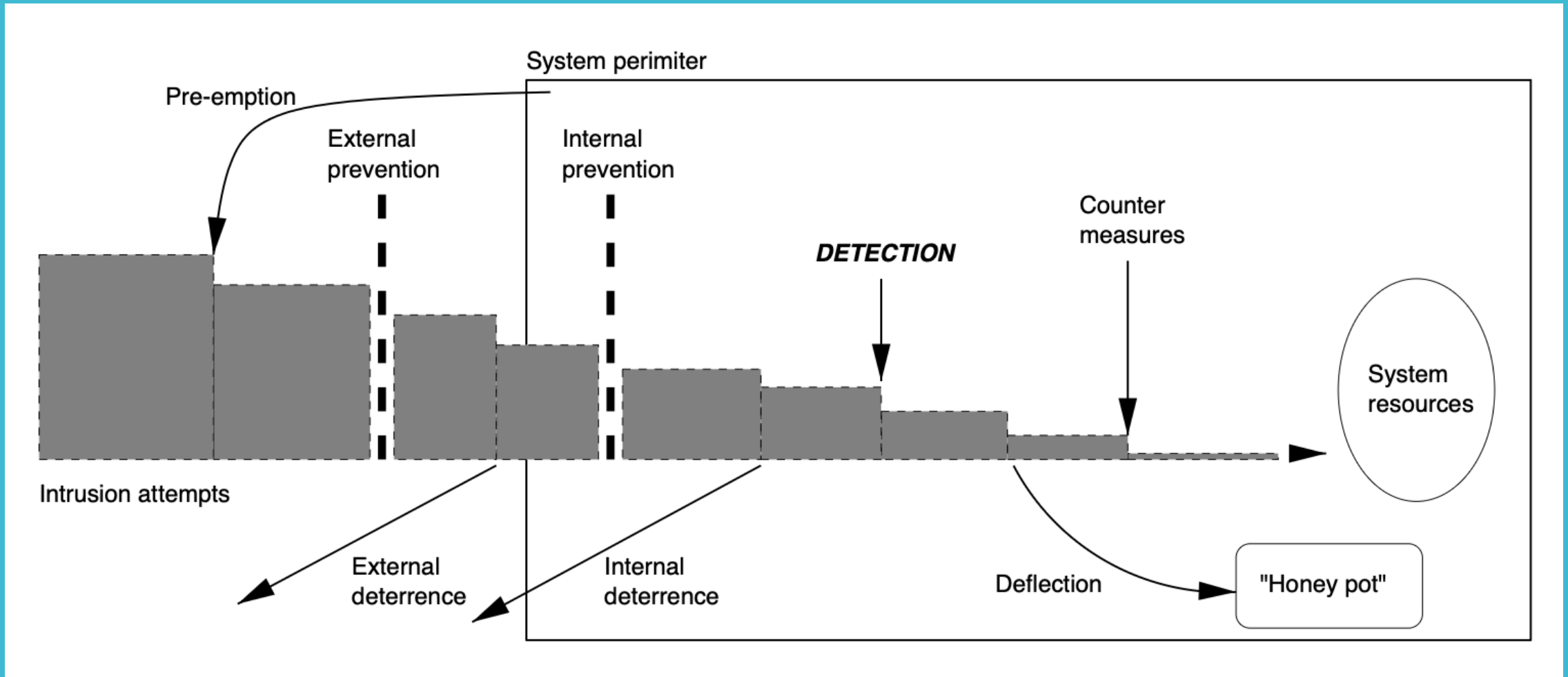
Information protection

Level	What to protect	Method
3	Existence of message	Steganography
2	Metadata of message	Privacy-enhancing technologies
1	Content of message	Encryption
0	Nothing	None

How to protect

- We have seen several approaches and techniques for the information protection
- Techniques were mainly focusing on how to make a security/privacy attack difficult
- Not all attacks may be prevented
- How to deal with the attacks anyway?

Defense lines: anti-intrusion methods





Pre-emption

Prevention

Deterrence

Detection

Deflection

Countermeasures

Taxonomy of anti-intrusion methods

Anti-intrusion methods

- **Pre-emption:** to strike against the threat before it could strike against us.
- **Prevention:** to preclude, or seriously reduce likelihood of a particular attack
 - It may be internal prevention, controlled by the system itself (system owner), or
 - It may be external, taking place in the environment of the system.
- **Deterrence:** to persuade an attacker not to launch an attack, or to stop ongoing attack. Usually done by increasing the risk of negative consequences for the attacker.

Anti-intrusion methods (cont.)



Detection: aims to find intrusion attempt and launch countermeasures



Deflection: to trick away an intruder from where he could do some damage ("honeypot" techniques)



Countermeasures: to actively counter an intrusion

Intrusion detection

- Intrusion detection is the most important of anti-intrusion methods:
 - Prevention, pre-emption and deterrence are not absolute, and attacks happen
 - For countermeasures one must detect an attack
- We consider general principles, structure and functionality of IDSs

Intrusion Detection System (IDS)

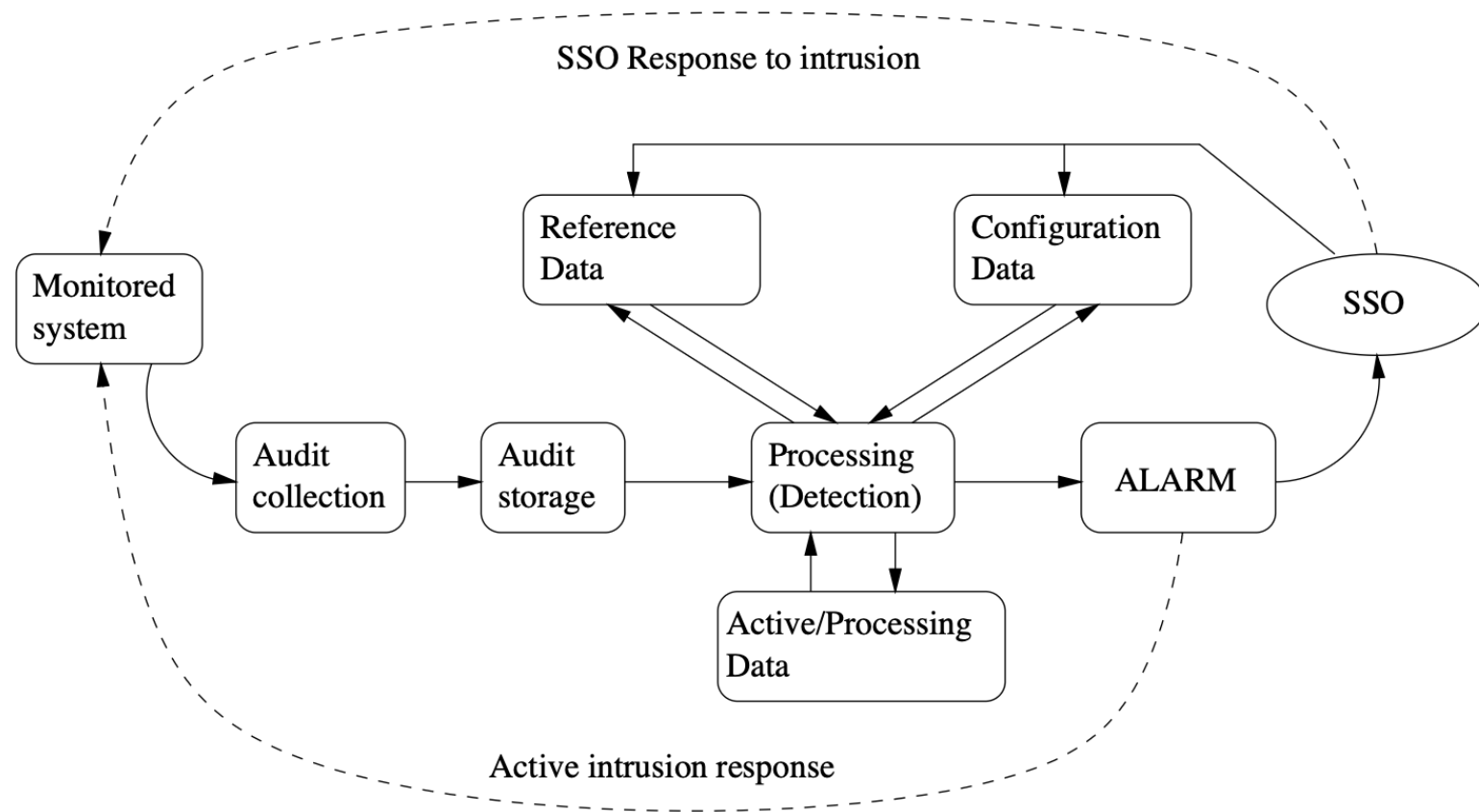
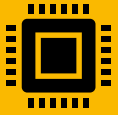


Figure 2: Organisation of a generalised intrusion detection system



Audit Collection: collect data for intrusion detection, including keyboard input, data from various log files, data on network activities



Audit Storage: stores the data for further processing, amount of data may be the problem



Processing: based on collected data, algorithm(s) are executed to find an evidence (with some degree of certainty) of the suspicious behaviour

Elements of typical IDS

Elements of typical IDS (cont.)

Configuration data: specify the way IDS works, how to collect data, how to respond to detected attack, etc

Reference data: information about known intrusion *signatures*, information about bad/normal behaviour

Active/Processing data: intermediate results, which should be stored during processing

Alarm

Types of IDS

1. Network-based IDS:

- Monitor network backbones
- Distributed among different nodes in the network
- Usually, passive => not easy to detect by an attacker
- May not be able analyse the traffic in large busy networks

2. Host-based IDS:

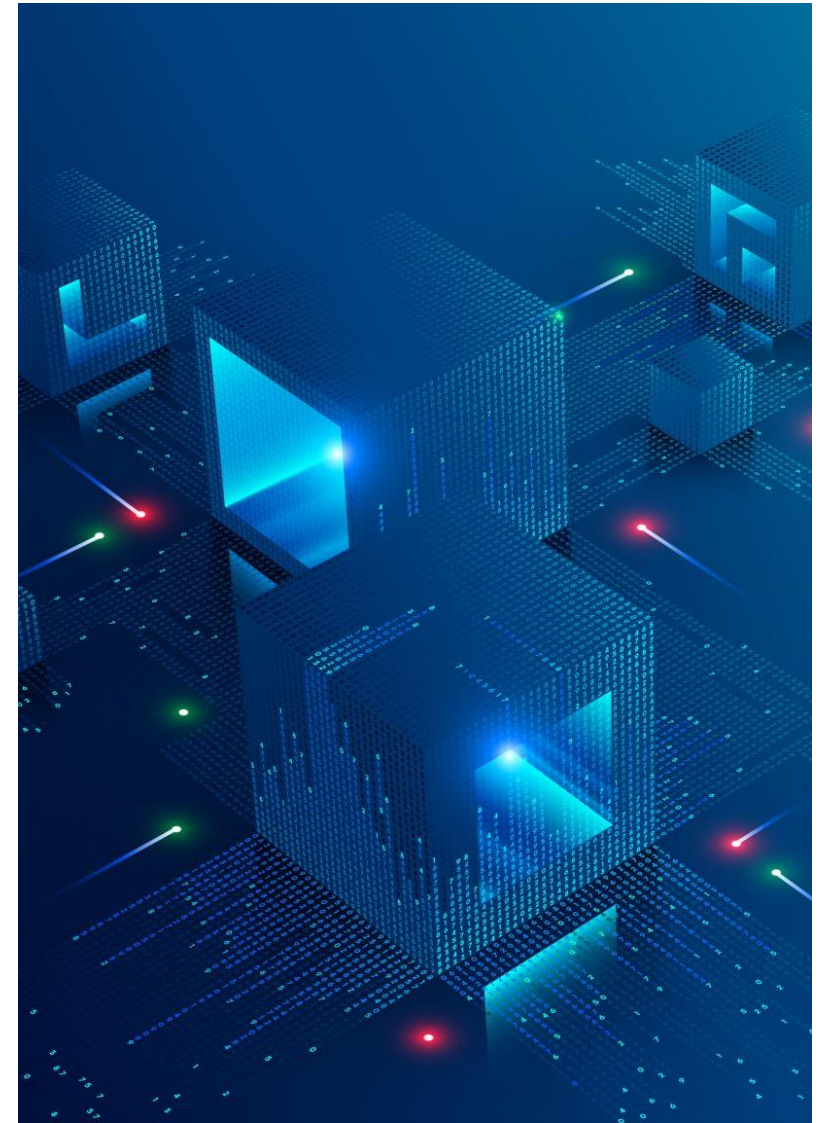
- Operate on hosts
- Defend and monitor the operating and file systems for signs of intrusion
- Usually monitor activities with higher level of details

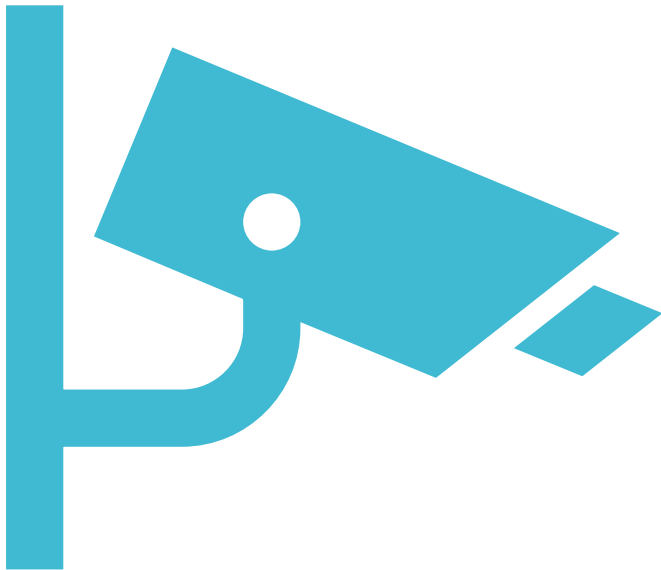
Types of IDS (cont.)

3. Application-based IDS:

Deal with the events appearing inside of a particular application, such as

- Database management systems
- Content management
- Accounting system





Intrusion detection methods

Two main categories:

- **Knowledge-based intrusion detection:** (policy-based, signature based, specification based): system tries to match the *explicit policies / signatures* with the data collected to find an evidence of the suspicious behaviour.
- **Anomaly-based intrusion detection:** system reacts to abnormal behaviour. Behaviour profiles are used, and system is able to learn what is “normal” behaviour.

Signature based method

Example of a signature: alert icmp \$EXTERNAL_NET any -> \$HOME_NET any (msg:"MISC large ICMP"; dsize: >800; reference:arachnids,246; classtype:bad-unknown; sid:499;)

Alarm will be raised if

- a ICMP packet originates from the external network,
 - can be associated with any port, and
 - the packet size is more than 800 bytes.
-
- A message appears in the IDS alert log.
 - Rest are metadata to track the triggering rule.

Signature based method

Advantages:

- Very low false alarm rate
- Simple algorithms, easy implementation

Disadvantages:

- Difficulties in updating information on new types of attacks
- Unable to detect unknown attacks (knowledge based)

Anomaly based detection

Advantages:

- Possibility of detection of novel attacks as intrusions
- Less dependence of IDSs on operating environment
- Ability to detect abuse of user privileges

Disadvantages:

- A substantial false alarm rate
- User behaviours can vary with time, requiring a constant update of the normal behaviour profile database.